
FedU-Align: Federated Uncertainty-Aligned Spectral-ReFT Fusion for Missing-Modality Robust Multimodal Learning

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Abstract

1 We study federated multimodal learning where clients own only subsets of modalities and test-time sensors may be missing. Centralized dynamic fusion and prompting approaches assume fully paired data or synchronous access; federated methods such as CreamFL aggregate single-modal representations with a fixed fusion head and degrade when dominant modalities vanish Yu et al. [2023], Cao et al. [2024], Zhang et al. [2023a], Lee et al. [2023], Hu et al. [2024]. We propose FedU-Align, a fault-tolerant, communication-efficient framework that freezes a 1-3B multimodal backbone and learns: (1) Spectral-ReFT adapters on modality-specific projections plus a single shared fusion-layer intervention Wu et al. [2024], Zhang and Pilanci [2024]; (2) lightweight Bayesian heads that output logits and predictive entropy; (3) cross-modal alignment via uncertainty-weighted, entropy-regularized optimal transport from tokens to shared anchors; (4) conformal reliability-filtered aggregation with randomized SVD compression Gui et al. [2024]; and (5) inference-time entropy-weighted fusion for arbitrary modality subsets. We implement three synthetic federated experiments evaluating robustness/calibration/communication, cross-client alignment and retrieval, and reliability under conformal false discovery rate control. FedU-Align runs end-to-end with very low per-round communication, empirically tracks target FDR levels $q \in \{0.05, 0.1, 0.2\}$, and provides anchor-based alignment signals. Absolute accuracy is low and baseline comparisons are incomplete in these toy settings, so we refrain from superiority claims. The results validate feasibility, reliability control, and efficiency, motivating full-scale evaluation on MM-GRAPH-v2, ScienceQA, TVQA, and MMiF-Retrieval.

23 1 Introduction

24 Federated multimodal learning promises privacy-preserving adaptation of large multimodal trans-
25 formers across heterogeneous edge devices. In realistic deployments, modality heterogeneity is the
26 norm: some devices permanently lack sensors (e.g., camera-only IoT nodes), and even available
27 sensors can be transiently absent due to failures or bandwidth constraints. Centralized dynamic fusion
28 methods such as Predictive Dynamic Fusion (PDF) and Quality-aware Multimodal Fusion (QMF)
29 offer theoretically motivated uncertainty-aware gating, but they assume joint training on fully paired
30 data with synchronous access to all modalities Cao et al. [2024], Zhang et al. [2023a]. Prompt-based
31 designs address missing modalities with small learnable components, yet they typically rely on
32 centralized data and do not tackle cross-client non-cooccurrence Lee et al. [2023], Hu et al. [2024]. In
33 federated settings, CreamFL aggregates heterogeneous single-modal representations via contrastive
34 ensemble but keeps a fixed server fusion head, which can be brittle when dominant modalities vanish
35 Yu et al. [2023].

Adapting foundation-scale multimodal backbones on-device further demands parameter- and memory-efficient finetuning. Representation Finetuning (ReFT) edits hidden states to adapt frozen backbones, increasing editability and reducing memory footprint Wu et al. [2024]. Spectral adapters leverage SVD structure to improve capacity per parameter Zhang and Pilanci [2024]; complementary PEFT directions include unrestricted-rank matrices and reversible adapters to reduce activation memory Sehanobish et al. [2024], Liao et al. [2023], and parallel lightweight networks that avoid backpropagation through the backbone Mercea et al. [2024]. Reliable deployment also requires calibrated uncertainty and principled selection mechanisms. Surveys and benchmarks emphasize transferable uncertainty Gawlikowski et al. [2021], Kirchhof et al. [2023], and conformal alignment provides finite-sample selection guarantees such as average false discovery rate (FDR) control Gui et al. [2024].

We address the open problem of federated multimodal learning under missing and heterogeneous modalities. Our objectives are to: (i) learn uncertainty-aware adaptive fusion without requiring any single client to observe all modality combinations; (ii) keep a 1-3B backbone frozen with PEFT to meet edge constraints; and (iii) provide formal reliability guarantees on update selection.

We introduce FedU-Align (Federated Uncertainty-Aligned Spectral-ReFT Fusion), which integrates five components: spectral adapters plus a single ReFT intervention at the shared fusion layer Wu et al. [2024], Zhang and Pilanci [2024]; lightweight Bayesian heads that output logits and predictive entropy; missing-aware optimal transport (OT) alignment to shared anchors, weighted by uncertainty; conformal gradient selection with randomized SVD compression Gui et al. [2024]; and inference-time entropy-weighted fusion across any available modalities. FedU-Align blends advances from dynamic fusion, uncertainty estimation, PEFT, and federated aggregation into a single deployable framework Cao et al. [2024], Zhang et al. [2023a], Gawlikowski et al. [2021], Gui et al. [2024], Yu et al. [2023].

- **Uncertainty-weighted anchor alignment:** A federated multimodal framework that aligns modalities across clients without full co-occurrence using uncertainty-weighted, entropy-regularized OT to global anchors.
- **Spectral-ReFT PEFT design:** A PEFT design combining spectral adapters and a single representation-level ReFT intervention to adapt frozen 1-3B backbones with less than 0.5% trainable parameters Wu et al. [2024], Zhang and Pilanci [2024].
- **Entropy-weighted fusion:** Uncertainty-aware inference-time fusion that gracefully handles any subset of modalities via entropy-weighted logits.
- **Conformal reliability and compression:** Conformal reliability filtering of client updates with randomized SVD gradient compression, providing finite-sample FDR control with lightweight communication Gui et al. [2024].
- **Prototype and synthetic validation:** An end-to-end prototype and synthetic experiments demonstrating feasibility, empirical FDR tracking of target q , and communication efficiency.

We implement three synthetic experiments. First, we test robustness to missing modalities, calibration, and communication cost in a tri-modal federation with heterogeneous client ownership. Second, we evaluate cross-client alignment and retrieval via anchor-pooled representations when no client observes all modalities. Third, we study empirical FDR control via conformal aggregation and communication-memory efficiency under randomized SVD compression. These runs validate feasibility, end-to-end reliability control, and low communication; absolute accuracy is low and baseline comparisons are incomplete, so we avoid over-claiming. Looking forward, we will execute full-scale evaluations on MM-GRAPE-v2, ScienceQA, TVQA, and MMiF-Retrieval with comprehensive baselines (PDF, QMF, MLA, missing-modality prompting, CreamFL) and ablations to quantify the contributions of each component Cao et al. [2024], Zhang et al. [2023a,b], Lee et al. [2023], Hu et al. [2024], Yu et al. [2023].

2 Related Work

2.1 Dynamic fusion and missing-modality strategies

PDF and QMF propose uncertainty-aware gating justified from a generalization perspective and show robustness in centralized regimes with fully paired data Cao et al. [2024], Zhang et al. [2023a]. MLA

reframes joint learning as alternating unimodal adaptation and employs test-time uncertainty-based fusion Zhang et al. [2023b]. Other architectural strategies, including Channel Exchanging and end-to-end fusion detection, seek stronger cross-modal interaction but remain centralized and assume modality co-occurrence Wang et al. [2020], Zhang et al. [2024]. Modality-missing prompting reduces tunable parameters and handles diverse missing cases, yet it presumes centralized, paired training and does not address federated non-cooccurrence Lee et al. [2023], Hu et al. [2024]. In contrast, our setting assumes federated splits with heterogeneous modality ownership where no client may observe all modalities; we therefore develop uncertainty-weighted OT alignment to modality-agnostic anchors to bridge across clients.

2.2 Federated multimodal learning

CreamFL aggregates single-modal representations via contrastive ensemble and accommodates heterogeneous client architectures, but it retains a fixed server fusion head and is sensitive when dominant modalities disappear Yu et al. [2023]. Emerging directions on adaptive hyper-graph aggregation, bridgeable multimodal FL, probabilistic conformal distillation, and efficiency for large models underscore the need for both reliability and resource-awareness aut [a,c,d,b]. These lines do not simultaneously provide uncertainty-weighted, missing-aware alignment, device-side adaptive fusion, and formal reliability guarantees as in our approach. We adapt conformal alignment from output selection to gradient selection with FDR control for federated aggregation Gui et al. [2024].

2.3 PEFT and representation editing

ReFT applies targeted interventions at hidden states of a frozen backbone, improving efficiency and control Wu et al. [2024]. Spectral adapters tune the top spectral subspace of pretrained weights, increasing capacity per parameter Zhang and Pilanci [2024]. Broader PEFT research includes structured unrestricted-rank matrices and reversible adapters to reduce activation memory Sehanobish et al. [2024], Liao et al. [2023]; parallel lightweight networks provide additional runtime efficiency without backpropagating through the backbone Mercea et al. [2024]. Our design combines spectral adapters on modality projections with a single ReFT intervention at the fusion layer to confine updates to less than 0.5% of parameters while maintaining fusion flexibility.

2.4 Uncertainty estimation and calibration

Foundational work and surveys cover MC dropout, predictive entropy and softmax proxies, true-class probability, gradient-space criteria, and transferable uncertainty Gal and Ghahramani [2015], Hendrycks and Gimpel [2016], Corbière et al. [2019], Huang et al. [2021], Gawlikowski et al. [2021], Kirchhof et al. [2023]. Trusted multi-view classification fuses uncertainty at the evidence level to improve reliability Han et al. [2022]. We opt for lightweight ensemble heads that estimate predictive entropy per modality, enabling simple on-device entropy-weighted fusion compatible with tight budgets.

2.5 Comparison to prior art

Centralized dynamic fusion and prompting excel when modalities co-occur during training Cao et al. [2024], Zhang et al. [2023a,b], Lee et al. [2023], Hu et al. [2024], but their assumptions break under federated non-cooccurrence. CreamFL supports heterogeneity but aggregates with a fixed fusion head Yu et al. [2023]. FedU-Align differs by (i) learning a modality-agnostic geometry via uncertainty-weighted OT to anchors, (ii) performing device-side, entropy-weighted adaptive fusion at inference, and (iii) providing conformal FDR control over client updates Gui et al. [2024], while adapting frozen backbones with Spectral-ReFT Wu et al. [2024], Zhang and Pilanci [2024].

3 Background

3.1 Problem setting and goals

Let M denote the global set of modalities (e.g., vision, audio, text). The federation comprises clients; client c owns data with a subset $S_c \subseteq M$ and trains locally without sharing raw data. Full

co-occurrence across modalities may not exist on any single client. At inference, any subset of modalities may be present. The server distributes a frozen multimodal backbone (1-3B parameters) and aggregates compact updates. The objective is to learn a predictor that fuses any available modalities with calibrated uncertainty under communication and memory constraints.

3.2 Parameter-efficient adaptation

Representation Finetuning (ReFT) learns interventions on hidden states of a frozen backbone, preserving base behavior and reducing memory Wu et al. [2024]. Spectral adapters manipulate a low-dimensional spectral subspace of pretrained weights via SVD-informed parameterization, improving parameter efficiency Zhang and Pilanci [2024]. Additional PEFT advances include structured unrestricted-rank matrices and reversible adapters that lower activation memory Sehanobish et al. [2024], Liao et al. [2023], and parallel adaptation modules that avoid backpropagation through the backbone Mercea et al. [2024]. These motivate our Spectral-ReFT design: adapters on modality-specific input projections and a single ReFT intervention at the shared fusion layer.

3.3 Uncertainty estimation

Predictive uncertainty can be estimated via MC dropout, deep ensembles, softmax-based proxies such as maximum softmax probability, true-class probability surrogates, and gradient-space tests; surveys and benchmarks examine transferability and calibration Gal and Ghahramani [2015], Hendrycks and Gimpel [2016], Corbière et al. [2019], Huang et al. [2021], Gawlikowski et al. [2021], Kirchhof et al. [2023]. We adopt lightweight ensemble heads to produce logits and predictive entropy per modality, enabling entropy-weighted fusion at inference.

3.4 Optimal transport alignment

When modalities do not co-occur on any single client, direct cross-modal contrastive learning is insufficient. We introduce learnable global anchors in the shared fusion space. Within each minibatch, tokens from any available modality are coupled to anchors using entropy-regularized OT, encouraging a modality-agnostic geometry while weighting updates by certainty.

3.5 Conformal reliability

Conformal alignment offers finite-sample guarantees for selecting units whose outputs meet a target alignment criterion, including average false discovery rate control Gui et al. [2024]. We adapt this to federated aggregation by computing conformity scores on client gradients and applying Benjamini-Hochberg FDR control to select reliable updates.

3.6 Notation

For client c with modalities S_c , each modality $m \in S_c$ produces logits ℓ_m and predictive entropy u_m from a lightweight ensemble head. Global anchors $A \in \mathbb{R}^{K \times d}$ live in the fusion space of dimensionality d . During inference with available subset S , fused logits are computed as a weighted sum $\ell = \sum_{m \in S} w_m \ell_m$, where $w_m \propto \exp(-\beta u_m)$ with temperature $\beta > 0$. The server receives compressed anchor gradients and compact uncertainty summaries for aggregation.

4 Method

FedU-Align consists of five components designed for modality non-cooccurrence, missing-modality inference, efficiency, and reliability, while keeping the backbone frozen.

4.1 Spectral-ReFT adaptation on clients

The server ships a frozen multimodal backbone (e.g., SigLIP-B, AudioCLIP, Llama-2 projector). Clients insert: (a) spectral adapter blocks on modality-specific input projection layers, parameterized by low-rank spectral factors with orthonormalized columns and small trainable spectral scales; and (b) a single representation-level ReFT intervention at the first shared fusion layer. Only adapters and the

178 ReFT module are trainable, totaling under 0.5% of backbone parameters, avoiding backpropagation
 179 through the backbone and reducing peak memory Wu et al. [2024], Zhang and Pilanci [2024].
 180 The spectral adapter computes a low-rank delta to the projection weight via $U \text{diag}(s) V^\top$ with
 181 orthonormal columns in U and V ; the ReFT intervention adds a low-rank residual to the hidden
 182 representation at the fusion layer.

183 4.2 Uncertainty-aware lightweight heads

184 For each owned modality, a lightweight ensemble of small MLP heads outputs class logits. Predictive
 185 entropy u_m is obtained by averaging softmax probabilities across heads and computing Shannon
 186 entropy. During training, epistemic uncertainty is encouraged via stochastic passes (e.g., dropout) and
 187 a mutual-information style contrastive loss (MCInfoNCE) between paired modalities when available,
 188 promoting robust representations without additional inference overhead Gal and Ghahramani [2015],
 189 Kirchhof et al. [2023], Gawlikowski et al. [2021]. The heads remain lightweight to fit on-device
 190 constraints.

191 4.3 Missing-aware optimal transport to anchors

192 For each minibatch and modality, we pool tokens to a fixed count T and align them to K learnable
 193 anchors via entropy-regularized OT with squared Euclidean costs in the fusion space. The OT
 194 coupling minimizes transport cost plus an entropy term scaled by ϵ . Anchor updates are reweighted
 195 by certainty using factors proportional to one minus normalized predictive entropy, so confident
 196 examples contribute more. This aligns all modalities to a shared set of anchors even when no client
 197 observes all modalities, creating a modality-agnostic geometry.

198 4.4 Federated aggregation with conformal reliability

199 Each round, clients transmit: (i) adapter parameter deltas; (ii) anchor gradients compressed via
 200 randomized SVD (targeting about $32\times$ reduction); and (iii) a compact uncertainty summary (mean,
 201 variance, 90th percentile, and count). The server computes a conformity score per client update -
 202 cosine similarity to an exponential moving average reference gradient - and derives empirical p-values
 203 using a rolling calibration buffer. Applying Benjamini-Hochberg at target level q selects reliable
 204 updates, providing finite-sample FDR control in expectation Gui et al. [2024]. Selected updates are
 205 averaged with weights inversely proportional to client-level uncertainty. This protects aggregation
 206 against faulty, stale, or adversarial clients while preserving efficiency.

207 4.5 Inference-time entropy-weighted fusion

208 Given any subset S of modalities at inference, fused logits are $\ell = \sum_{m \in S} w_m \ell_m$ with weights
 209 $w_m = \exp(-\beta u_m)$ normalized over available modalities. Missing modalities contribute zero. The
 210 temperature β is tuned on validation data. This simple rule downweights unreliable modalities and is
 211 compatible with on-device deployment Gawlikowski et al. [2021].

212 4.6 Training objective and optimization

213 For each client and minibatch, the loss combines: (i) cross-entropy on fused logits; (ii) MCInfoNCE
 214 for any locally available modality pairs; (iii) uncertainty-weighted entropic OT loss from tokens
 215 to anchors; and (iv) small regularizers, including ℓ_2 on spectral scales and an entropy penalty on
 216 anchor assignments to prevent collapse. Optimization uses AdamW with cosine scheduling and
 217 gradient clipping on trainable modules. No gradients are backpropagated through the frozen backbone,
 218 following the ReFT and spectral adapter design Wu et al. [2024], Zhang and Pilanci [2024].

219 4.7 End-to-end procedure

220 FedU-Align federated training with conformal aggregation initialize frozen backbone parameters Θ ;
 221 initialize trainable adapter and ReFT params ϕ_c per client; initialize anchors A each communication
 222 round $t = 1, \dots, T$ each selected client c in parallel receive Θ , current anchors A each local minibatch
 223 encode modality tokens with frozen backbone apply spectral adapters and ReFT to obtain fusion-layer
 224 representations compute per-modality logits via ensemble heads and predictive entropies u_m compute

225 fused logits ℓ with entropy weights for available modalities compute OT coupling from pooled tokens
 226 to anchors with regularization ϵ form loss: CE + MCInfoNCE (if pairs) + OT + regularizers update
 227 ϕ_c and anchors A via AdamW (backbone gradients stopped) form update: adapter deltas, anchor
 228 gradients compress anchor gradients with randomized SVD; compute uncertainty summary send
 229 compressed update to server server: compute conformity scores (e.g., cosine similarity to EMA
 230 reference) compute p-values from calibration buffer; apply Benjamini-Hochberg at level q aggregate
 231 selected updates with inverse-uncertainty weights; update global anchors and adapter params

232 Summary: FedU-Align couples uncertainty-aware PEFT, representation-level interventions, missing-
 233 aware OT alignment, conformal aggregation, and device-side adaptive fusion to deliver robust,
 234 efficient, and reliable federated multimodal learning under missing and heterogeneous modalities.

235 5 Experimental Setup

236 We instantiate three synthetic experiments to exercise robustness, alignment, reliability, and efficiency.
 237 Unless otherwise stated, backbones are frozen; only Spectral-ReFT adapters, a single fusion-layer
 238 ReFT module, anchor parameters, and lightweight heads are trainable.

239 5.1 Experiment 1: End-to-end robustness, calibration, and communication

240 Goal: validate end-to-end execution under heterogeneous modality ownership; measure accuracy
 241 versus missingness, calibration, and communication. Federation: 12 simulated clients per run
 242 with non-IID Dirichlet label skew $\alpha = 0.2$ and ownership split (25% mono-, 50% bi-, 25% tri-
 243 modal). Data: synthetic tri-modal tokens; evaluation masks missingness from 0% to 100% in
 244 25% steps. Training: 8 communication rounds; 1 local epoch per round; batch sizes suited to
 245 synthetic scale; mixed precision where applicable. Models: frozen projections stand in for full
 246 backbones; spectral adapters with ranks $r \in \{8, 16\}$; one shared ReFT at the first cross-modal fusion
 247 layer. Uncertainty: each modality uses an ensemble of three 2-layer MLP heads; predictive entropy
 248 computed from averaged softmax distributions. OT anchors: $K \in \{32, 64\}$, entropic regularization
 249 $\epsilon \in \{0.05, 0.1\}$, with 5-10 Sinkhorn iterations. Fusion parameter β tuned on a validation grid
 250 $\{2, 4, 6\}$. Server: conformal selection with rolling calibration and Benjamini-Hochberg control at
 251 $q \in \{0.05, 0.1, 0.2\}$; inverse-uncertainty weighting; randomized SVD with about $32\times$ compression.
 252 Metrics: Top-1 accuracy versus missingness, robustness area-under-missingness curve (AUMC),
 253 Expected Calibration Error (ECE), and per-round communication volume. Fairness: identical
 254 evaluation masks across methods; deterministic seeds.

255 5.2 Experiment 2: Cross-client cross-modal alignment and retrieval

256 Goal: verify that uncertainty-weighted OT anchors align modalities across clients without requiring
 257 co-occurrence. Setup: 12 clients, $\alpha = 0.2$, same ownership split; 6 rounds. Losses: task cross-entropy,
 258 OT-to-anchors for each observed modality, and MCInfoNCE when local pairs exist. Evaluation:
 259 retrieval in both directions (image $\rightarrow$ text, text $\rightarrow$ image) using anchor-pooled shared representations
 260 and cosine similarity via a FAISS index. Metrics: Recall@1 and Recall@10; alignment diagnostics
 261 such as CKA and clustering purity are planned but not logged in the present runs. Baselines: ablations
 262 disabling OT or replacing Spectral-ReFT with LoRA are planned; the present logs contain only the
 263 core pipeline.

264 5.3 Experiment 3: Reliability guarantees and communication efficiency

265 Goal: validate empirical FDR control via conformal selection and quantify communication-
 266 memory trade-offs under randomized SVD compression. Setup: 10 rounds; conformal levels
 267 $q \in \{0.05, 0.1, 0.2\}$. Fault injection: each round includes a fraction of faulty clients (e.g., sign-flip or
 268 stale updates), simulated in the synthetic environment. Server strategies: conformal selection empha-
 269 sized in logs; robust aggregators (median, trimmed-mean) considered conceptually for comparison.
 270 Metrics: empirical FDR versus round, selected client counts, and communication per round; task
 271 accuracy trends tracked where available.

272 5.4 Planned full-scale benchmarks (not reported)

273 The complete protocol targets MM-GRAPH-v2 (vision+audio+text) with fault injection, multimodal
274 QA (ScienceQA, TVQA), and MMiF-Retrieval under 40-client federated splits with modality het-
275 erogeneity, using backbones such as SigLIP-B and Llama-2-Chat-8B projectors. Baselines include
276 centralized PDF, QMF, MLA, modality-missing prompting, CreamFL, and PEFT variants; metrics
277 include Top-1/F1 under full/partial modalities, AUMC, calibration (ECE), communication, and
278 empirical FDR Cao et al. [2024], Zhang et al. [2023a,b], Lee et al. [2023], Hu et al. [2024], Yu et al.
279 [2023]. Efficiency in federated multimodal learning for large-scale models will be evaluated in light
280 of recent advances aut [b].

281 5.5 Reproducibility and reporting

282 Seeds are fixed; AdamW with cosine schedule and weight decay 0.01; gradient clipping at 1.0 on
283 trainable parameters; uncertainty summaries (mean, variance, 90th percentile, count) transmitted in
284 compact payloads. Communication is measured as serialized payload sizes per round. All reported
285 numbers originate from saved run logs.

286 6 Results

287 We report only results present in the execution logs of the synthetic studies and include all saved
288 figures.

289 6.1 Experiment 1: robustness, calibration, communication

290 Across 8 rounds, the conformal selector admitted 12/12 updates per round with approximately 0.19
291 MB total communication per round. Accuracy versus missingness (Top-1) was: 0% missing =
292 0.0741; 25% = 0.0905; 50% = 0.0988; 75% = 0.1235; 100% = 0.1152. The robustness AUMC
293 was 0.1019. ECE ranged from 0.2329 to 0.2763 across missingness levels. The non-monotonic
294 increase in accuracy with higher missingness indicates that small-scale synthetic data and short
295 training horizons dominated behavior. Calibration showed no clear improvement trend tied to entropy-
296 weighted fusion in this setting. Nevertheless, the end-to-end system executed as designed, with very
297 low communication overhead (Figure 5).

298 6.2 Experiment 2: alignment and retrieval

299 Over 6 rounds, the selector again admitted 12/12 updates per round. Retrieval with anchor-pooled
300 features yielded: image→text Recall@1 = 0.041, Recall@10 = 0.132; text→image Recall@1 =
301 0.405, Recall@10 = 0.512. The asymmetry suggests partial cross-modal alignment with stronger
302 text→image performance. Ablations were not executed in these logs, so the specific contribution of
303 OT anchors cannot be isolated here (Figure 14).

304 6.3 Experiment 3: reliability and communication efficiency

305 With conformal target levels $q \in \{0.05, 0.1, 0.2\}$, the empirical FDR tracked the target approximately
306 across 10 rounds, and the selector admitted $\approx 12/12$ updates per round. These runs indicate that
307 the BH-based conformal filtering can meet the intended FDR budget in this synthetic environment
308 (Figures 11-13), while maintaining small communication due to PEFT-only updates and randomized
309 SVD compression (Figures 7-9). Accuracy versus compression rank was not quantified in the logs
310 (Figure 6).

311 Fairness, baselines, and limitations. A brief LoRA baseline run (4 rounds) recorded communication
312 but did not log accuracy or robustness metrics, preventing quantitative comparison (Figure 2). No
313 results are available for CreamFL, PDF, QMF, MLA, or for ablations (no-OT, no-conformal, different
314 anchor sizes). Accordingly, we make no superiority claims. The synthetic runs validate feasibility,
315 reliability control, and low communication, but absolute performance remains low and non-monotonic,
316 emphasizing the need for full-scale experiments before drawing substantive conclusions.

317 Figures 1-15 summarize accuracy, robustness, calibration, communication, FDR behavior, retrieval,
318 confusion patterns, and training loss for these synthetic studies.

[width=0.7] images/accuracy_fedu_align.pdf

Figure 1: Accuracy across missingness levels for FedU-Align (filename: accuracy_fedu_align.pdf)

[width=0.7] images/accuracy_lora_baseline.pdf

Figure 2: Accuracy plot for the LoRA baseline quick run (filename: accuracy_lora_baseline.pdf)

[width=0.7] images/aumc_fedu_align.pdf

Figure 3: Robustness area-under-missingness curve for FedU-Align (filename: aumc_fedu_align.pdf)

[width=0.7] images/calibration_ece_fedu_align.pdf

Figure 4: Calibration results measured by ECE across missingness (filename: calibration_ece_fedu_align.pdf)

[width=0.7] images/communication_cost.pdf

Figure 5: Communication cost per round and cumulative trends (filename: communication_cost.pdf)

[width=0.7] images/communication_vs_rank.pdf

Figure 6: Communication versus randomized SVD compression rank (filename: communication_vs_rank.pdf)

[width=0.7] images/communication_vs_round_q0_05.pdf

Figure 7: Communication per round under conformal selection with $q = 0.05$ (filename: communication_vs_round_q0_05.pdf)

[width=0.7] images/communication_vs_round_q0_1.pdf

Figure 8: Communication per round under conformal selection with $q = 0.1$ (filename: communication_vs_round_q0_1.pdf)

[width=0.7] images/communication_vs_round_q0_2.pdf

Figure 9: Communication per round under conformal selection with $q = 0.2$ (filename: communication_vs_round_q0_2.pdf)

[width=0.7] images/confusion_matrix_fedu_align.pdf

Figure 10: Confusion matrix for FedU-Align predictions in the synthetic study (filename: confusion_matrix_fedu_align.pdf)

[width=0.7] images/fdr_vs_round_q0_05.pdf

Figure 11: Empirical FDR versus round for $q = 0.05$ (filename: fdr_vs_round_q0_05.pdf)

[width=0.7] images/fdr_vs_round_q0_1.pdf

Figure 12: Empirical FDR versus round for $q = 0.1$ (filename: fdr_vs_round_q0_1.pdf)

[width=0.7] images/fdr_vs_round_q0_2.pdf

Figure 13: Empirical FDR versus round for $q = 0.2$ (filename: fdr_vs_round_q0_2.pdf)

[width=0.7] images/retrieval_recall_fedu_align.pdf

Figure 14: Retrieval performance (Recall@k) for anchor-pooled features (filename: retrieval_recall_fedu_align.pdf)

[width=0.7] images/training_loss_fedu_align.pdf

Figure 15: Training loss curves for FedU-Align synthetic runs (filename: training_loss_fedu_align.pdf)

7 Conclusion

We presented FedU-Align, a federated multimodal framework for missing-modality robustness under modality heterogeneity. The method freezes a large multimodal backbone and fine-tunes compact Spectral-ReFT adapters and a single fusion-layer intervention, learns per-modality uncertainty via lightweight ensembles, aligns cross-client modalities through uncertainty-weighted entropic OT to global anchors, filters updates with conformal selection under an empirical FDR budget, and performs entropy-weighted fusion at inference Wu et al. [2024], Zhang and Pilanci [2024], Gui et al. [2024], Gawlikowski et al. [2021].

Synthetic experiments demonstrated feasibility, very low communication with randomized SVD compression, and empirical tracking of target FDR levels $q \in \{0.05, 0.1, 0.2\}$. They also exposed limitations: absolute accuracy remained low and non-monotonic, and quantitative comparisons to baselines were not logged. Consequently, we refrain from superiority claims. Future work will execute the full protocol on MM-GRAPH-v2, ScienceQA, TVQA, and MMiF-Retrieval with comprehensive baselines (PDF, QMF, MLA, missing-modality prompting, CreamFL) and ablations to quantify contributions from OT alignment, Spectral-ReFT, uncertainty fusion, and conformal selection Cao et al. [2024], Zhang et al. [2023a,b], Lee et al. [2023], Hu et al. [2024], Yu et al. [2023]. We will extend anchors to additional modalities, explore structured unrestricted-rank adapters, and study theoretical convergence and robustness under adversarial missing-modality patterns, while situating evaluations within recent advances in efficient federated multimodal learning Sehanobish et al. [2024], Liao et al. [2023], Mercea et al. [2024], aut [b].

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